Solution Requirements

Date	4 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Secure user access via email/password or SSO integration for policy analysts.*	**High**
2	Authorization levels	*Two roles: 'Admin' (can update ML models) and 'User' (can run predictions).*	**High**
3	External interfaces	*Integration with UNDP/World Bank Open Data APIs for automated dataset fetching.*	**Medium**

4	Transactions processing	*Handling input of numerical socio-economic data and outputting a predicted HDI score.*	**High**
5	Reporting	*Dashboard generation showing HDI score, visual trend charts, and export to PDF/CSV.*	**High**
6	Business rules, etc.	*System must validate that all 4 input fields are within historically realistic ranges (e.g., GNI > 0).*	**High**
7	Compliance to laws or regulations	*Adherence to data privacy standards (GDPR) regarding the storage of user input queries.*	**High**
8	*Other*	*Language toggle (English/Regional) for global accessibility.*	**Low**

<u>Step 2: Non-Functional Requirements (NFR):</u>

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	*API response time for HDI prediction must be instantaneous.*	*Loads in < 2 seconds*

2	Scalability	*System should handle multiple users running simultaneous simulations.*	*Supports 500+ concurrent users*
3	Security & Data Privacy	*All data transmitted between the web UI and Flask API must be secure.*	*HTTPS/TLS 1.2+ encryption*
4	Reliability & Availability	*The system hosted on Render must be available consistently for daily analysis.*	*99.5% Uptime*
5	Usability & Accessibility	*Intuitive dashboard design using simple sliders and clear input labels.*	*WCAG 2.1 AA Compliance*
6	*Other*	*Codebase must be modular and documented for easy updates to ML models.*	Maintainable/Scalable